



**ALLIANCE
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**NAAC
GRADE A+**
ACCREDITED UNIVERSITY

Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 1

**Title: Importing business datasets from flat files & MySQL and
connecting with any other sources into Power BI**

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Importing business datasets from flat files & MySQL and connecting with any other sources into Power BI

GitHub-Repo-Link: <https://github.com/keerthi19hub/Business-Analytics>

Objective and Problem Statement

The objective of this experiment was to connect Power BI to a MySQL retail database, prepare and transform the business data, build a related data model, create reusable DAX measures, and develop an interactive three-page dashboard for sales, customer, store, payment, and demographic analysis.

The retail company data contains transactional sales information together with product, store, and customer details. The final report provides management with a centralized view of business performance and customer behaviour.

MySQL Database Creation and SQL Operations

The source data was first prepared in MySQL Workbench. A database named RetailDB was created and selected using CREATE DATABASE RetailDB and USE RetailDB. The Products, Stores, and Sales tables were then created with suitable primary keys, data types, and transaction fields, and business records were inserted using INSERT INTO statements.

The Sales table acts as the main transaction table and contains SaleID, ProductID, StoreID, Quantity, SaleDate, and TotalAmount. Products and Stores provide descriptive information that can be related to the sales transactions.

Connecting MySQL to Power BI and Importing Data

Power BI Desktop was connected to the MySQL RetailDB instance through the Get Data option. The retaildb sales, retaildb products, and retaildb stores tables were selected from the database and loaded through Power Query. A Customer Details table and a Calendar table were also included to support customer analysis and time-based analysis.

- Opened Power BI Desktop and selected Get Data.
- Connected to the MySQL RetailDB database.
- Selected the required retail tables from the database.
- Loaded Sales, Products, Stores, Customer Details, and Calendar data.
- Opened Power Query Editor to prepare the imported datasets for analysis.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- moviestreamingdb
- my_project
- normalization
- normalization_db5rf
- normalizationdb
- normalizationdb4rf
- normalizationdb5rf
- normalizationdb
- retaildb**
 - Tables
 - Views
 - Stored Procedures
 - Functions
- smartmoviedb
- smartmoviestreamingdb
- smartmoviestreamingdb2
- students
- sys**

Administration Schemas

Information

Schema: sys

SQL File 14"

Limit to 1000 rows

```

1 CREATE DATABASE RetailDB;
2 USE RetailDB;
3
4 -- 1. Products Table
5 CREATE TABLE Products (
6     ProductID INT PRIMARY KEY,
7     ProductName VARCHAR(100),
8     Category VARCHAR(50),
9     Price DECIMAL(10,2),
10    Supplier VARCHAR(100)
11 );
12
13 -- Insert Products
14 INSERT INTO Products
15 (ProductID, ProductName, Category, Price, Supplier)
16 VALUES
17 (201, 'Tablet', 'Electronics', 22000.00, 'Apple'),
18 (202, 'Smartwatch', 'Electronics', 12000.00, 'Fitbit'),
19 (203, 'Desk Lamp', 'Furniture', 1200.00, 'Philips'),
20 (204, 'Wireless Mouse', 'Accessories', 900.00, 'HP'),
21 (205, 'Bookshelf', 'Furniture', 4500.00, 'IKEA');

```

Output

#	Time	Action	Message	Duration / Fetch
1	16:35:21	CREATE DATABASE RetailDB	1 row(s) affected	0.016 sec
2	16:35:21	USE RetailDB	0 row(s) affected	0.000 sec
3	16:35:21	CREATE TABLE Products (ProductID INT PRIMARY KEY, ProductName VARCHAR(100), Category V...	0 row(s) affected	0.015 sec
4	16:35:21	INSERT INTO Products (ProductID, ProductName, Category, Price, Supplier) VALUES (201, 'Tablet', 'Electronic...	7 row(s) affected Records: 7 Duplicates: 0 Warnings: 0	0.016 sec
5	16:35:22	CREATE TABLE Stores (StoreID INT PRIMARY KEY, StoreName VARCHAR(100), City VARCHAR(50)...	0 row(s) affected	0.016 sec
6	16:35:22	INSERT INTO Stores (StoreID, StoreName, City, State, StoreType) VALUES (11, 'Kolkata Hub', 'Kolkata', 'West...	6 row(s) affected Records: 6 Duplicates: 0 Warnings: 0	0.000 sec

Context Help Snippets

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Insert Modeling View Optimize Help

Get data - workbook catalog - Server data - Data

Transform Refresh data - Queries

New visual

Text box More visuals -

New visual calculation - measure me

Calculators

MySQL database

Server

localhost:3306

Database

retaildb

Advanced options

OK Cancel

Import data from

Get data from another source ->

Navigator

Display Options

- localhost:3306: retaildb [3]
 - retaildb.products
 - retaildb.sales
 - retaildb.stores

No items selected for preview

Select Related Tables

Load Transform Data Cancel

Data Cleaning and Transformation

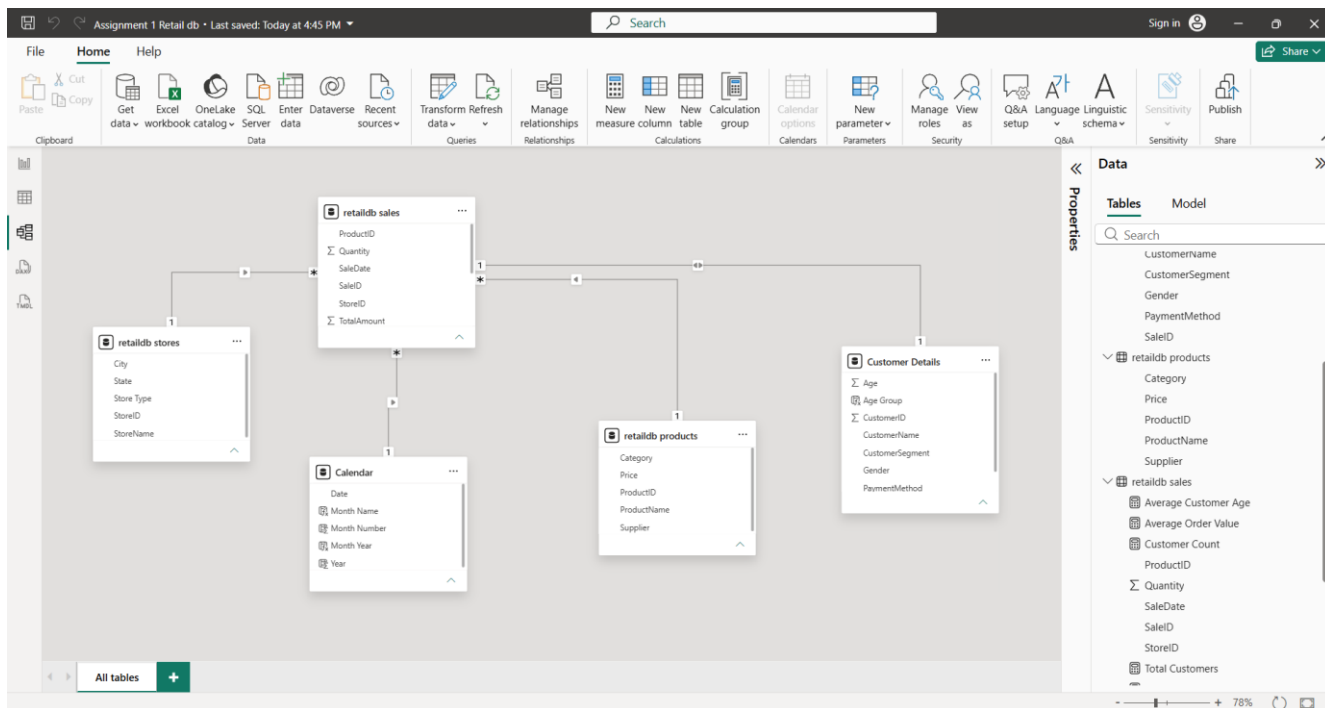
After importing the data, the datasets were cleaned and shaped in Power Query Editor. The purpose of this stage was to make the data consistent, analysis-ready, and suitable for relationships and calculations in Power BI.

Cleaning / Transformation	What was done
Duplicate records	Removed duplicate records from the sales and customer datasets.
Missing values	Checked key fields for blank or missing values and handled them where required.
Column naming	Renamed columns where necessary to maintain clear and consistent business-friendly names.
Data types	Converted SaleDate to Date, TotalAmount and Price to Decimal, and Quantity/IDs to appropriate numeric types.
Age Group	Created an Age Group field with 18–25, 26–35, and 36–45 groups for demographic analysis.
Calendar fields	Created Month Name, Month Number, Month Year, and Year from the Date field.

Data Modeling

A star-schema style model was created in Power BI Model view. The retaildb sales table was used as the central fact table, while retaildb products, retaildb stores, Customer Details, and Calendar were used as supporting dimension tables. The relationships allow filters such as City, Category, Month, Gender, or Payment Method to flow into the visuals.

- retaildb sales → retaildb products using ProductID.
- retaildb sales → retaildb stores using StoreID.
- retaildb sales → Customer Details using SaleID.
- retaildb sales → Calendar using the date field.
- Cardinality and cross-filter direction were checked to avoid ambiguous filtering.



DAX Measures Created

Reusable DAX measures were created so that KPI cards and charts respond dynamically to slicers and filters. The following measures were used in the report.

Measure	DAX Formula	Purpose
Total Sales	Total Sales = SUM(retaildb_sales[TotalAmount])	Calculates total revenue generated across all transactions.
Total Orders	Total Orders = DISTINCTCOUNT(retaildb_sales[SalesID])	Counts unique sales transactions/orders.
Total Quantity	Total Quantity = SUM(retaildb_sales[Quantity])	Calculates the total number of units sold.
Total Customers	Total Customers = DISTINCTCOUNT('Customer Details'[CustomerID])	Counts unique customers who made a purchase.
Average Order Value	Average Order Value = DIVIDE([Total Sales], [Total Orders], 0)	Calculates the average revenue generated per order.

Average Customer Age	Average Customer Age = AVERAGE('Customer Details'[Age])	Calculates the average age of customers.
Customer Count	Customer Count = COUNTROWS('Customer Details')	Returns the total number of customer records in the model.

<<

Filters

Data

...

>>

Search

Calendar

Date

Month Name

Month Number

Month Year

Year

Customer Details

Age

Age Group

CustomerID

CustomerName

CustomerSeg...

Gender

PaymentMeth...

SaleID

retaildb products

Category

Price

ProductID

ProductName

Supplier

retaildb sales

Average Cust...

Average Orde...

Customer Cou...

ProductID

Quantity

SaleDate

SaleID

StoreID

Total Customers

Total Orders

Total Quantity

Total Sales

TotalAmount

retaildb stores

City

State

Store Type

StoreID

StoreName

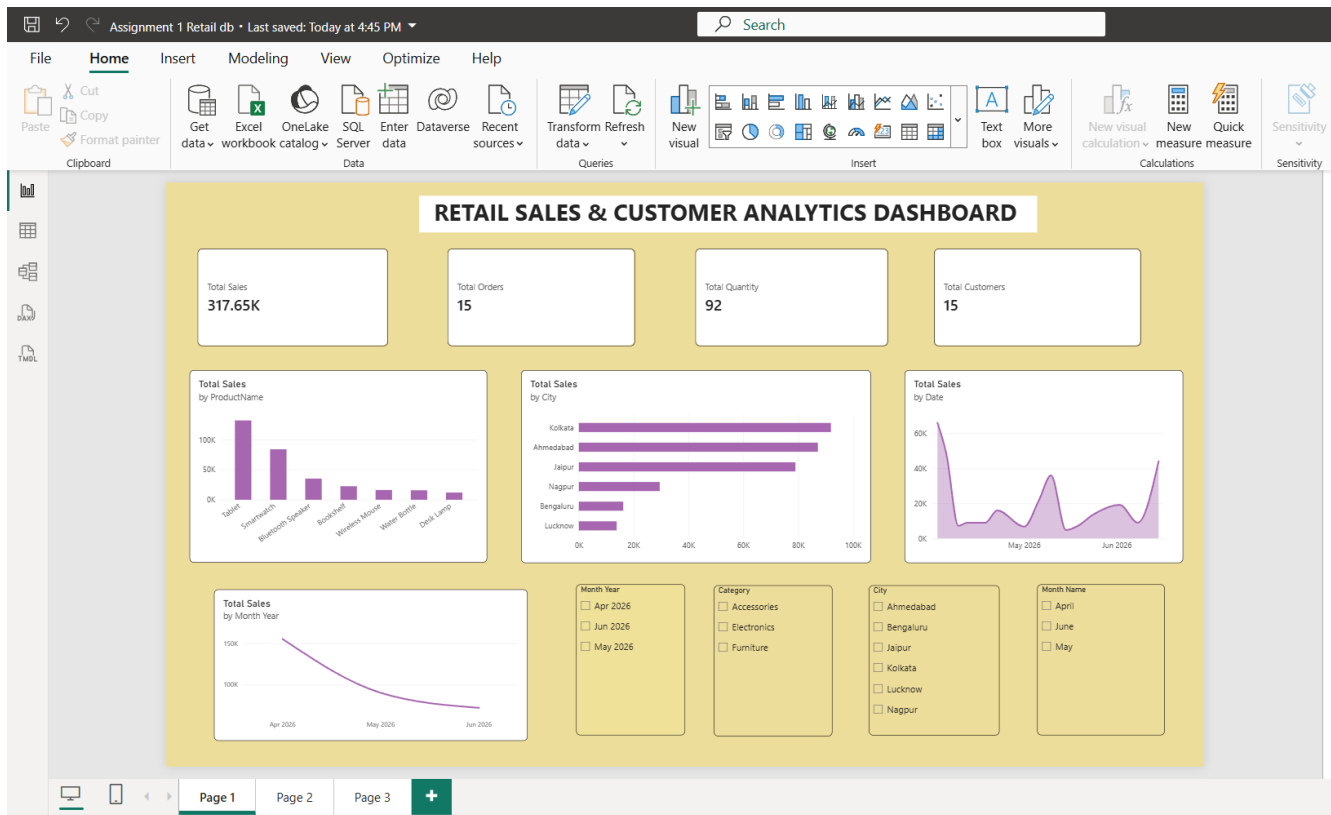
Dashboard Development and Explanation

The final Power BI report contains three dashboard pages. Each page focuses on a different analytical perspective while using the same underlying data model and DAX measures.

Page 1 – Retail Sales & Customer Analytics Overview

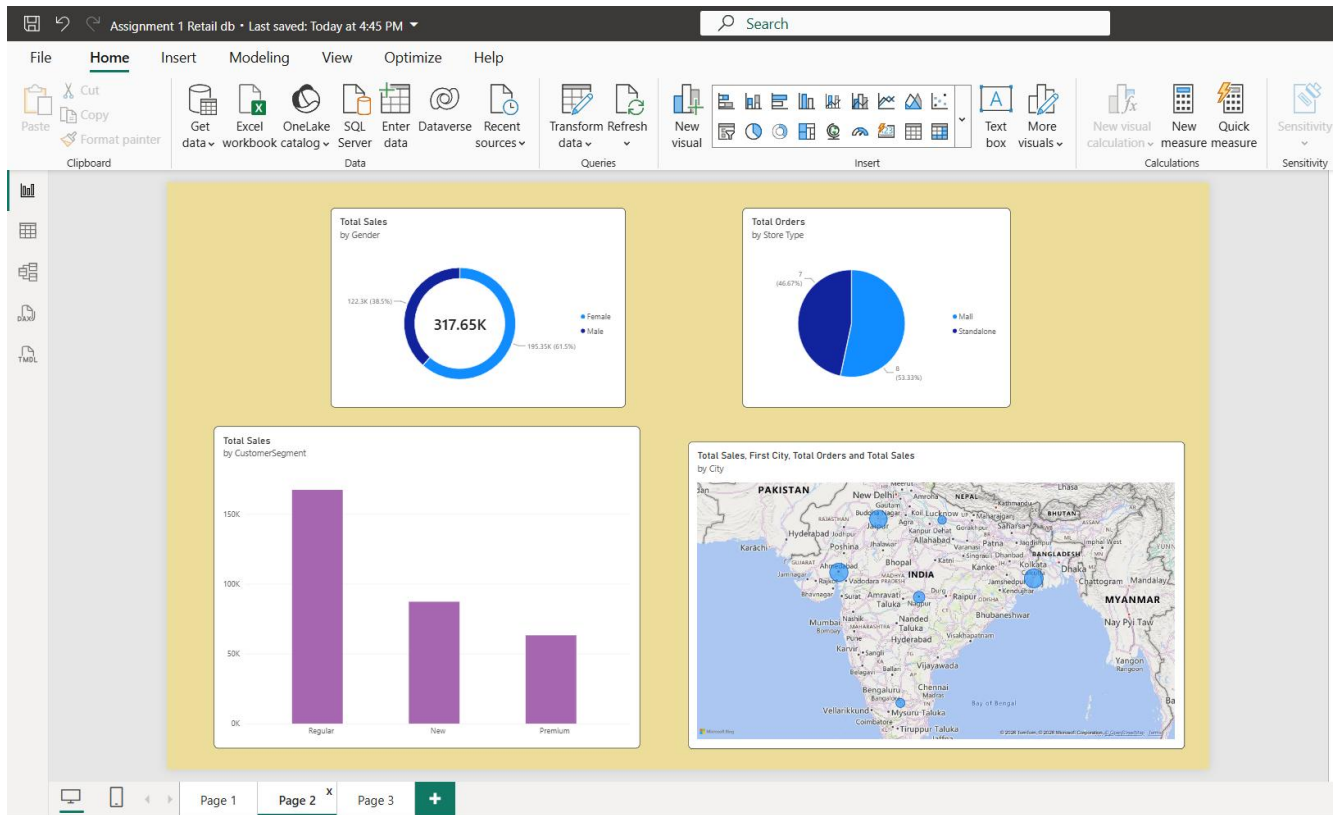
The first page is the main overview dashboard. It provides four KPI cards showing Total Sales (317.65K), Total Orders (15), Total Quantity (92), and Total Customers (15). It also contains product-wise sales, city-wise sales, sales by date, and sales by month-year visuals.

The Month Year, Category, City, and Month Name slicers make the page interactive. When a user selects a filter, the KPI cards and charts update according to the selected data.



Page 2 – Customer & Store Segmentation

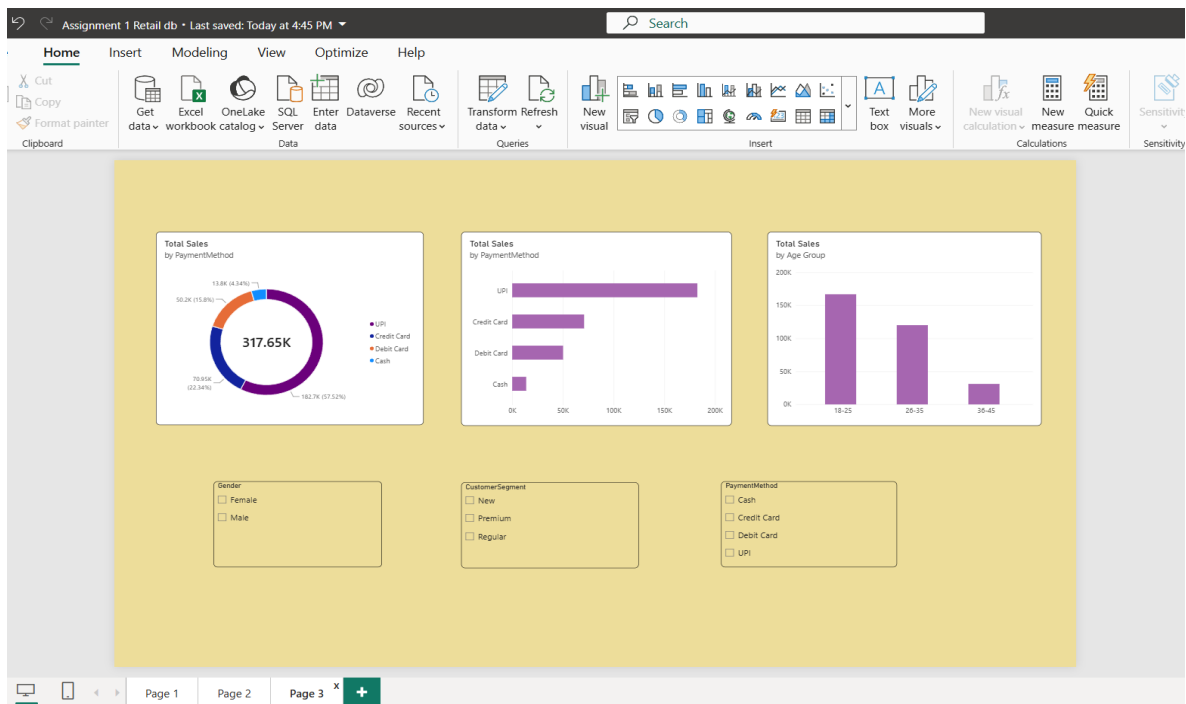
The second page focuses on customer and store segmentation. A donut chart compares Total Sales by Gender, a pie chart shows Total Orders by Store Type, and a column chart compares Total Sales across Regular, New, and Premium customer segments.



Page 3 – Payment & Demographic Analysis

The third page analyses payment behaviour and age demographics. Total Sales is shown by PaymentMethod using UPI, Credit Card, Debit Card, and Cash. A separate bar chart provides the same payment comparison in an easy-to-rank format.

The Age Group chart compares sales for the 18–25, 26–35, and 36–45 groups. Gender, CustomerSegment, and PaymentMethod slicers provide additional interactive filtering.



Conclusion

The experiment successfully connected Power BI to the MySQL RetailDB source, imported the required business datasets, cleaned and transformed the data, and created a related data model. DAX measures were then developed to power the KPI cards and analytical visuals.

The resulting three-page dashboard provides a centralized and interactive view of retail sales, customer behaviour, store performance, payment methods, and demographic trends. The report can be used to explore business performance through slicers and visual comparisons.